Experiment 20

K-means clustering algorithm for a given data set

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline
```

✓ 0.0s

```
df=pd.read_csv('Mall_Customers.csv')
```

✓ 0.0s

```
df.info()
```

✓ 0.0s

```
<class 'pandas.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   CustomerID              200 non-null    int64
 1   Gender                  200 non-null    object
 2   Age                     200 non-null    int64
 3   Annual Income (k$)      200 non-null    int64
 4   Spending Score (1-100)  200 non-null    int64
dtypes: int64(4), object(1)
memory usage: 7.9+ KB
```

```
df.head()
```

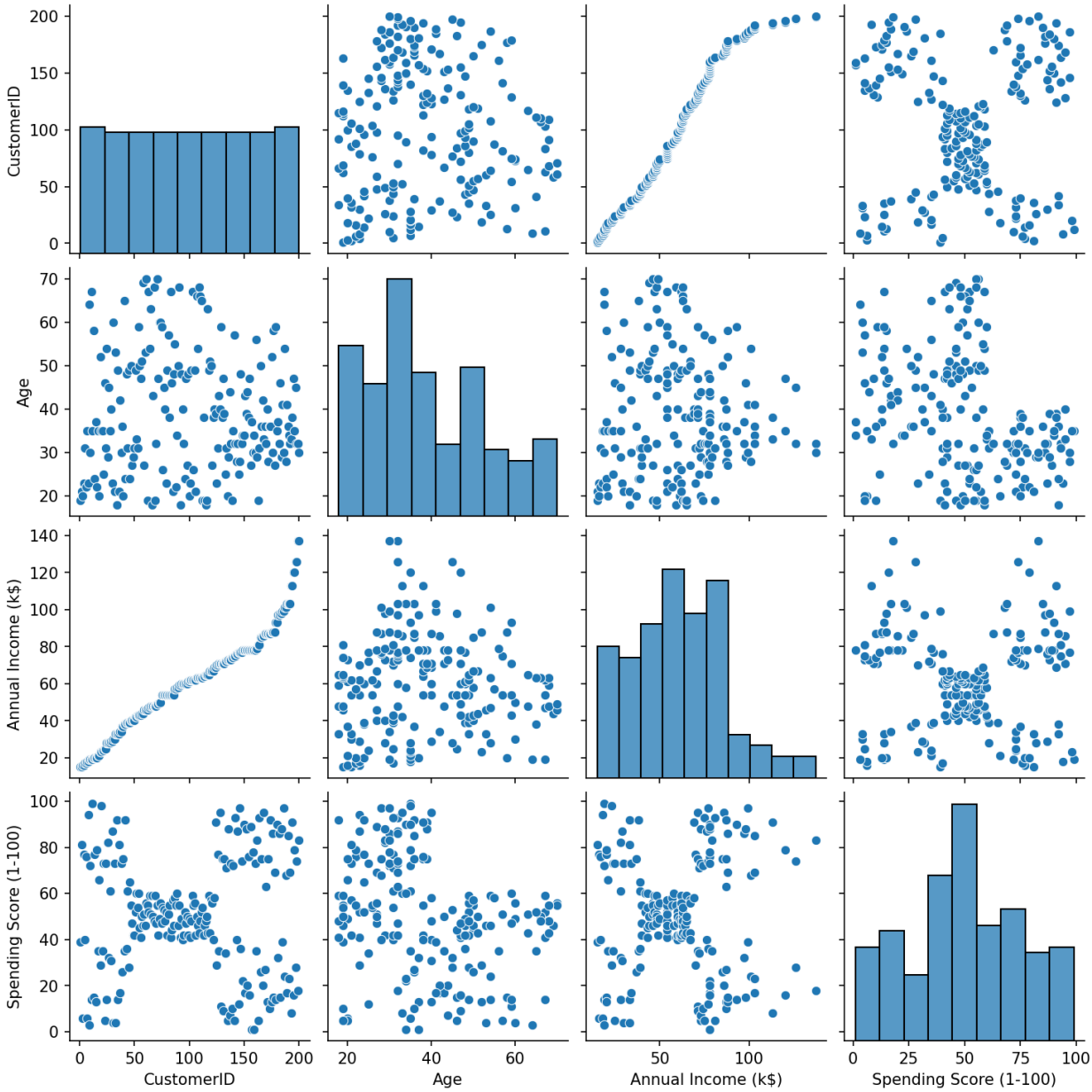
✓ 0.0s

	CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

```
sns.pairplot(df)
```

✓ 2.8s

```
<seaborn.axisgrid.PairGrid object at 0x7f6f3591b500>
```




```
features=df.iloc[:,[3,4]].values
```

✓ 0.0s

```
from sklearn.cluster import KMeans
model=KMeans(n_clusters=5)
model.fit(features)
KMeans(n_clusters=5)
```

✓ 0.4s

```
KMeans(n_clusters=5)
```

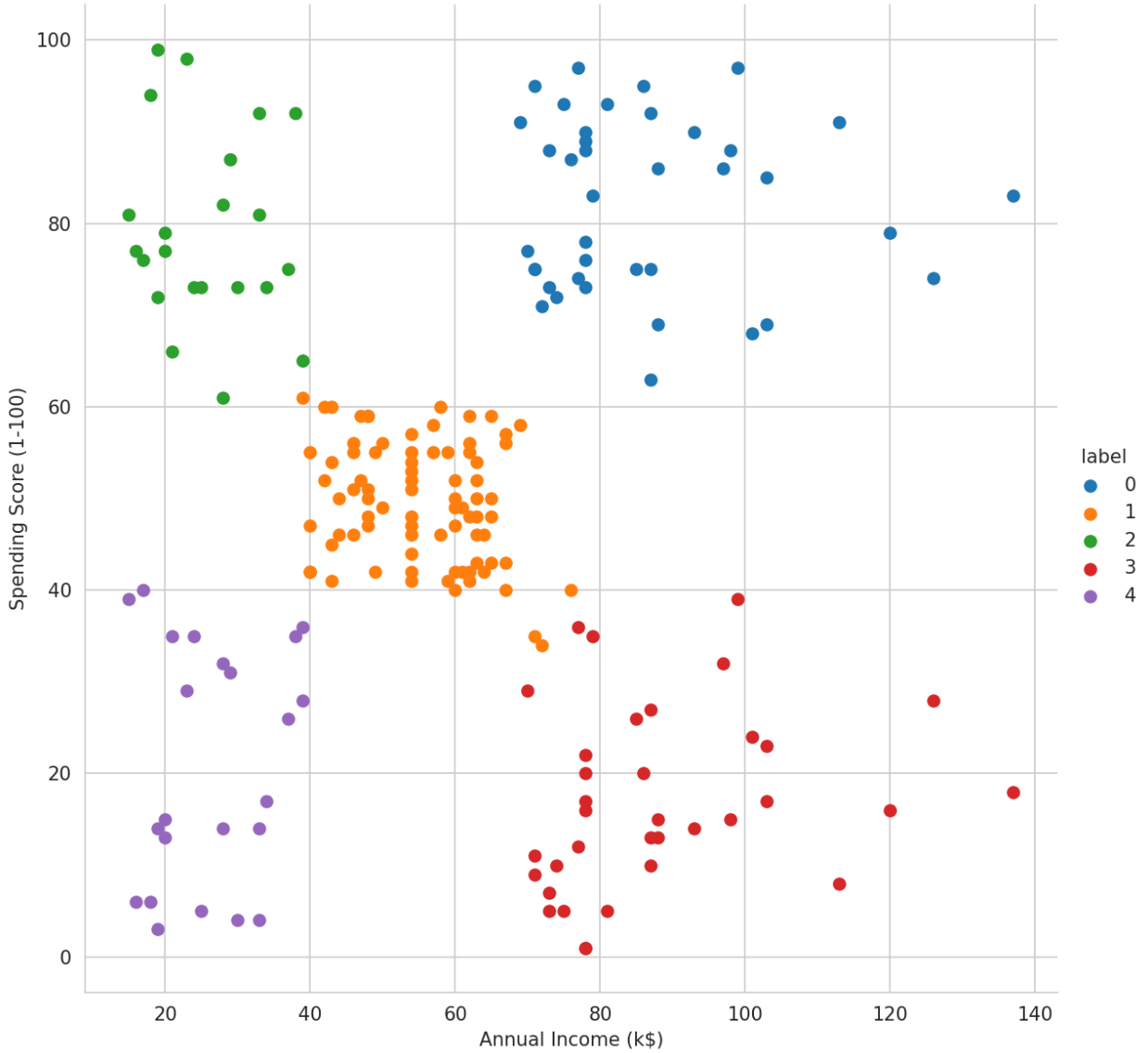
```
Final=df.iloc[:,[3,4]]
Final['label']=model.predict(features)
Final.head()
```

✓ 0.0s

	Annual Income (k$)	Spending Score (1-100)	label
0	15	39	4
1	15	81	2
2	16	6	4
3	16	77	2
4	17	40	4

```
sns.set_style("whitegrid")
sns.FacetGrid(Final,hue="label",height=8) \
.map(plt.scatter,"Annual Income (k$)", "Spending Score (1-100)") \
.add_legend();
plt.show()
```

✓ 0.5s



```
features_el=df.iloc[:,[2,3,4]].values
from sklearn.cluster import KMeans
wcss=[]
for i in range(1,10):
  model=KMeans(n_clusters=i)
  model.fit(features_el)
  wcss.append(model.inertia_)
plt.plot(range(1,10),wcss)
```

✓ 0.1s

```
[<matplotlib.lines.Line2D object at 0x7f6f2f447530>]
```

